

E-Commerce US Dataset Analysis

Report 6: Feature Engineering

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1. Objective

- Transform cleaned tables into model-ready features for downstream ML tasks.
- Build four separate feature tables: master_features, customer_features, product_features, seller_features.
- All features are derived entirely from existing columns — no external data added.

2. Order-Level Features (master_features.csv)

2.1 Delivery Timing Features

Feature	Derivation	Type
delivery_days	order_delivered_customer_date – order_purchase_timestamp (days)	Numeric
estimated_days	order_estimated_delivery_date – order_purchase_timestamp (days)	Numeric
delivery_delay	delivery_days – estimated_days (negative = early)	Numeric
is_late	1 if delivery_delay > 0, else 0	Binary
delivery_speed	Fast (<5 days) / Medium (5–10) / Slow (>10)	Categorical

2.2 Payment Behavior Features

Feature	Derivation	Type
avg_item_value	total_price / n_items	Numeric
freight_ratio	total_freight / total_price	Numeric
uses_installments	1 if max_installments > 1, else 0	Binary
installment_category	Single (1) / Short-term (2–4) / Long-term (5–12)	Categorical
is_high_value	1 if total_payment > \$3,857.70 (IQR upper fence)	Binary

2.3 Profit Features

- profit: total_price minus total cost of items in the order.
- margin_pct: profit as a percentage of total_price.

3. Customer-Level Features (customer_features.csv — 7,898 rows)

3.1 Transactional Features

Feature	Description
total_orders	Count of orders placed by the customer
total_spend	Sum of all payment amounts
avg_order_value	Mean order size (total_spend / total_orders)
total_items	Cumulative items purchased across all orders

3.2 Lifecycle Features

Feature	Description
first_order	Timestamp of the customer's earliest order
last_order	Timestamp of the customer's most recent order
tenure_days	Days between first and last order (0 for single-order customers)
is_repeat_customer	1 if total_orders > 1, else 0

3.3 Engagement Features

Feature	Description
review_rate	Proportion of orders where the customer submitted a review
avg_review_given	Mean review score given by the customer (1–5)
engagement_score	Weighted composite: 40% order frequency + 30% review rate + 30% normalized spend

3.4 Segmentation Feature

- value_tier: quartile-based spend segmentation — Bronze (Q1), Silver (Q2), Gold (Q3), Platinum (Q4).

4. Seller-Level Features (seller_features.csv — 500 rows)

Feature	Description
total_revenue	Sum of item prices for all items sold by the seller
total_items_sold	Count of items sold
avg_price	Mean item price across the seller's catalog
n_unique_orders	Number of distinct orders the seller contributed to
n_products	Number of distinct product_ids the seller listed
revenue_per_order	total_revenue / n_unique_orders
seller_tier	Tertile-based revenue segmentation: Low / Mid / Top

5. Product-Level Features (product_features.csv — 1,980 rows)

Feature	Description
times_ordered	How many times the product appeared in an order (demand frequency)
total_revenue	Sum of all revenue generated by this product
avg_price	Mean item price recorded for this product across orders
popularity_tier	Tertile-based demand: Low / Medium / High
product_revenue_rank	Descending rank by total_revenue (rank 1 = highest revenue product)

6. Feature Output Summary

File	Rows	Source Table	Location
master_features.csv	10,000	master_clean.csv	data/processed-feature/
customer_features.csv	7,898	master_clean.csv	data/processed-feature/
product_features.csv	1,980	items_clean.csv	data/processed-feature/
seller_features.csv	500	items_clean.csv	data/processed-feature/

7. Validation Results

- A DataValidator class with 9 validation methods was applied to master_features.csv.

Validation Check	Rule	Status
Schema completeness	All 39 expected columns present	PASS
total_payment dtype	Column is float64	PASS
n_items dtype	Column is int64	PASS
customer_age dtype	Column is int64	PASS
Missing value threshold	No column exceeds 10% null rate	PASS
Payment rule	total_payment >= 0 for all rows	PASS
Age rule	customer_age in [18, 100]	PASS
Item count rule	n_items >= 0 for all rows	PASS
Referential integrity	All customer_unique_id in customer_table	PASS
City-state consistency	Each city maps to exactly one state	PASS

- All 10 validation checks passed. Validation results saved to reports/Validation_report/master_validation.csv.